9/7 ch.1.1

* $2x+3y+5z=5$
- varibles: x, y, z 變數
- coefficient: 2, 3, 5 係數
- costant term: 5
- equations: $2x+3y+5z=5$ 等式
- system: many linear equations

ex.
$$\begin{cases} a_1x+b_1y+c_1z=d_1 \\ a_2x+b_2y+c_2z=d_2 \\ \vdots \\ a_nx+b_ny+c_nz=d_n \end{cases}$$

* 符號
- 粗體小寫 vector (向量): $\vec{u}$, $\vec{v}$, $\vec{w}$ $\begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$, $[1, 3, -2]$
- 細體小寫: scalar (純量)
- 細體大寫: matrix (矩陣)

* type of solution:

1. unique solution
2. no solution
3. infinite solutions

9/8 ch1.1

part of a matrix:
element: the number in the macrix
row: horizontal line of number
column: vetical line of number
submatrix: 子矩陣

size of matrix:

2x3 matrix $\begin{bmatrix} 2 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix}$

2x2 matrix $\begin{bmatrix} 3 & 3 \\ 1 & 2 \end{bmatrix}$ square matrix

1x3 matrix $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ row matrix

3x1 matrix $\begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$ column matrix

identity matrix: 單位矩陣 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

a system of 3 varibles and 3 equations.

$$x_1+x_2+x_3=2$$
$$2x_1+3x_2+x_3=3$$
$$x_1-x_2-2x_3=-6$$

→ in matrix of coefficint

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & -1 & -2 \end{bmatrix}$$

(增廣矩陣)

→ in matrix of augmented

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$

row equivalent: Different augmented matrix but same solution.

符号: ≈

☆ Elementary Row Operation (ERO)

① Interchange

② Multiply

③ Add a multiple of the element of one row.